

LG Vehicles Report Workflow — User Guide

This guide walks you through configuring and running the LG Vehicles Report Workflow, which generates a vehicle movement analysis report for Lion Guardians in the Amboseli ecosystem sourced from EarthRanger.

Overview

The workflow produces, for each tracked vehicle:

- A **Speed Map** — trajectory segments coloured by a 6-class speed palette (green → red)
 - A **Vehicle Tracks Map** — uniform blue path overlay on boundary layers
 - A **Speed Over Time line chart** — speed (km/h) plotted against date per vehicle
 - **Scalar metric widgets** — Mean Speed, Min Speed, Max Speed, Distance covered
 - A **per-vehicle summary CSV** — speed and distance statistics with a totals row
 - A **Word document report** (`.docx`) — cover page plus one section per vehicle
 - An **interactive widget dashboard**
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Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance with a configured data source
- The **Vehicles** subject group present in your EarthRanger instance

The two spatial boundary files (group ranch boundaries and conflict hotspot areas) and both Word report templates are downloaded automatically from Dropbox — no local copies are required.

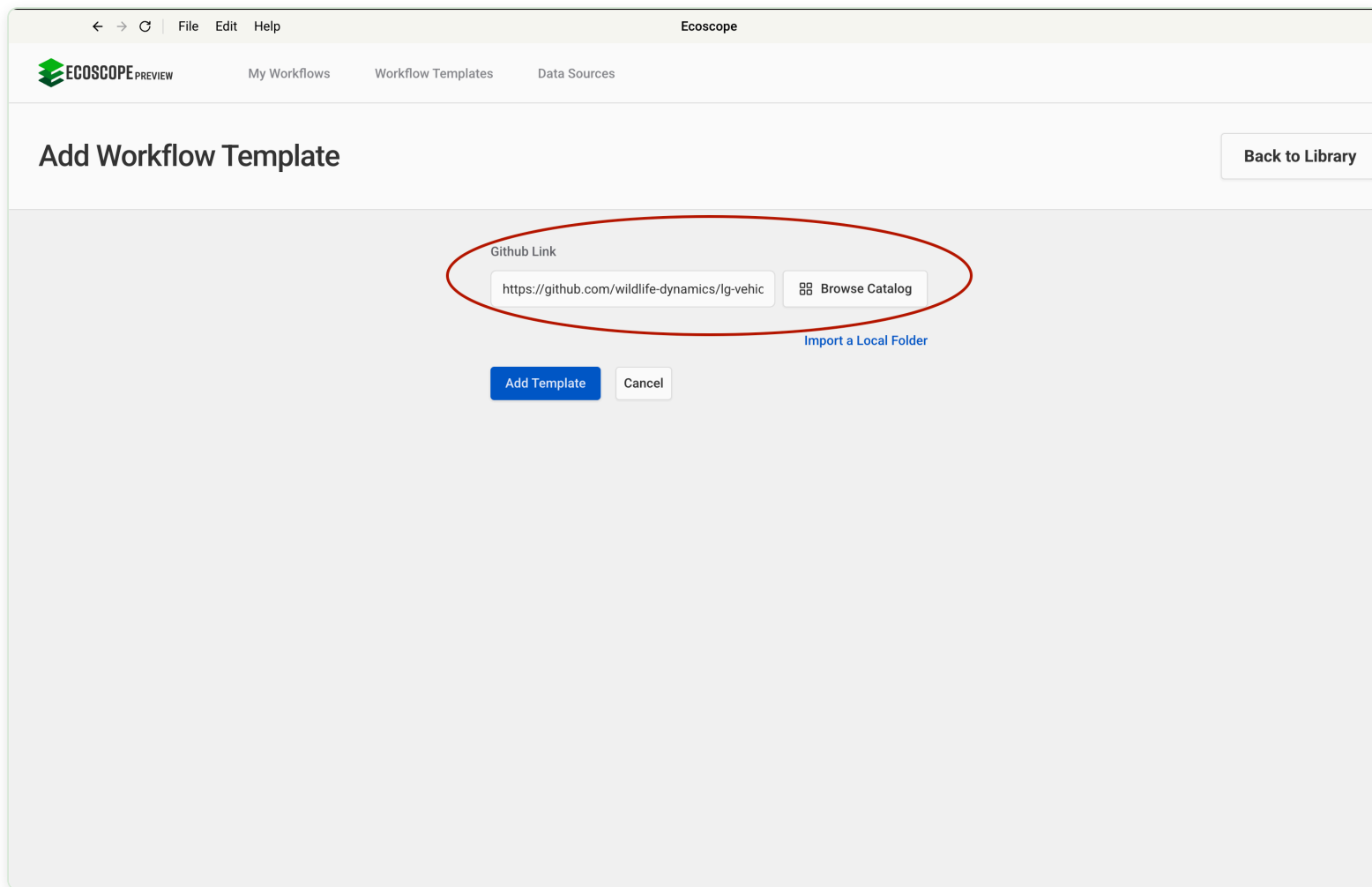
Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field:

```
https://github.com/wildlife-dynamics/lg-vehicles-report.git
```

Then click **Add Template**.



Add Workflow Template

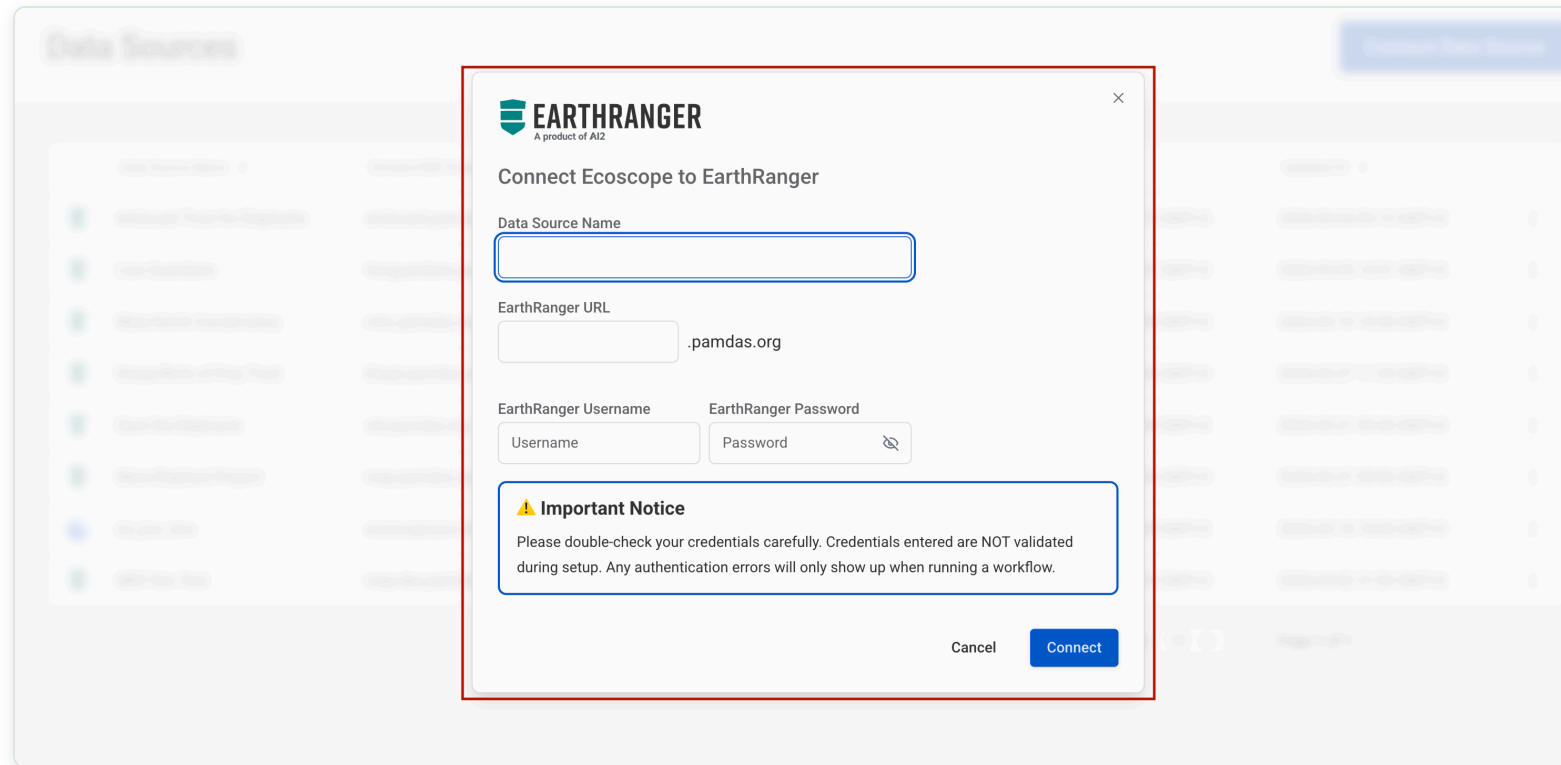
Step 2 — Add an EarthRanger Connection

Navigate to **Data Sources** and add a new EarthRanger connection. Fill in:

- **Data Source Name** — a label to identify this connection
- **EarthRanger URL** — your instance URL (e.g. `your-site.pandas.org`)

- **EarthRanger Username** and **EarthRanger Password**

Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

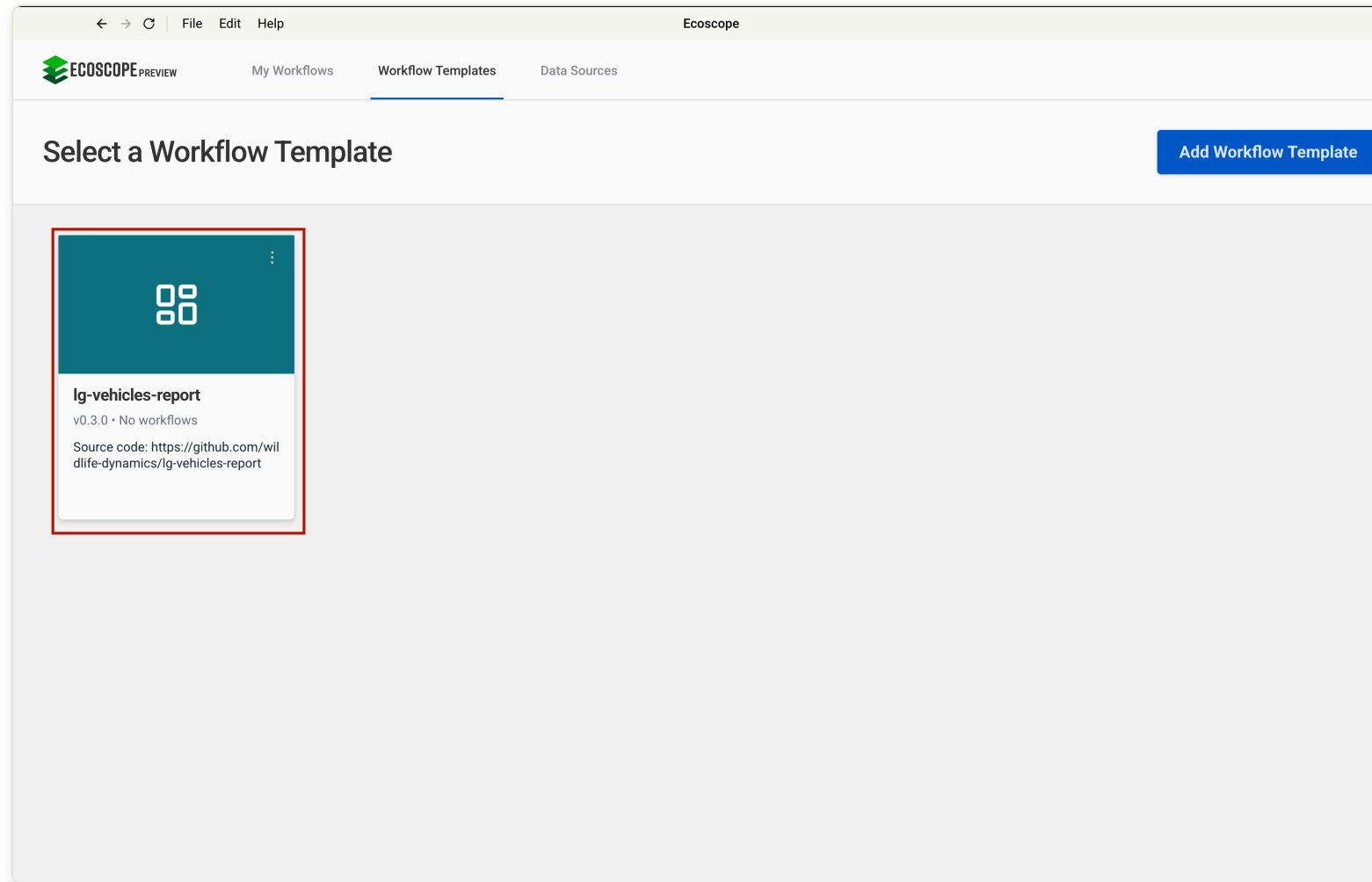
A screenshot of a web application interface showing a 'Data Sources' list in the background. Overlaid on this is a modal dialog box titled 'Connect Ecoscope to EarthRanger'. The dialog box has a red border and a close button (X) in the top right corner. It contains the following fields: 'Data Source Name' (a text input field), 'EarthRanger URL' (a text input field with '.pamdas.org' entered), 'EarthRanger Username' (a text input field with 'Username' entered), and 'EarthRanger Password' (a password input field with 'Password' entered and a toggle icon). Below these fields is a blue-bordered box with a yellow warning icon and the text: 'Important Notice: Please double-check your credentials carefully. Credentials entered are NOT validated during setup. Any authentication errors will only show up when running a workflow.' At the bottom right of the dialog are 'Cancel' and 'Connect' buttons.

EarthRanger Connection

Step 3 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **lg-vehicles-report**. Click it to open the workflow configuration form.

The card may show **Initializing...** briefly while the environment is set up.



Select Workflow Template

Step 4 — Set Workflow Details and Define Time Range

The configuration form opens with two sections at the top.

Set Workflow Details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes about the run (e.g. date range or vehicle group)

Define time range

Field	Description
Timezone	Select the local timezone (e.g. <code>Africa/Nairobi UTC+03:00</code>)
Since	Start date and time of the analysis period
Until	End date and time of the analysis period

All vehicle tracks and metrics are computed within this window.

Workflow Templates > Workflow Configuration

Set Workflow Details

Define time range

Data Source

Transform relocations to trajectories

zoom to envelope

Set Workflow Details

Add information that will help to differentiate this workflow from another.

Workflow Name*

Workflow Description

Define time range

Choose the period of time to analyze.

Timezone *

Africa/Nairobi (UTC+03:00)

Since *

05/03/2026, 12:00 AM

The start time

Until *

04/04/2026, 11:59 PM

The end time

Cancel Submit

Set Workflow Details and Define Time Range

Step 5 — Connect to EarthRanger, Transform Relocations to Trajectories, and Zoom to Envelope

Scroll down to configure the remaining three sections.

Connect to EarthRanger

Select the EarthRanger connection configured in Step 2 from the **Connect to EarthRanger** dropdown. The workflow will fetch all observations for the **Vehicles** subject group from this instance.

Transform relocations to trajectories *(Advanced Configurations)*

These parameters control how raw GPS fixes are converted into trajectory segments. Expand **Advanced Configurations** to adjust the trajectory segment filters:

Field	Default	Description
Minimum Segment Length (m)	3	Discard segments shorter than this distance
Maximum Segment Length (m)	100000	Discard segments longer than this distance
Minimum Segment Duration (s)	1	Discard segments shorter than this duration
Maximum Segment Duration (s)	21600	Discard segments longer than this duration (6 hours)
Minimum Segment Speed (km/h)	3	Discard segments below this average speed
Maximum Segment Speed (km/h)	150	Discard segments above this average speed

Zoom to envelope

Field	Default	Description
Expansion Factor	1.05	Factor to expand the bounding box when auto-zooming maps (e.g. 1.2 = 20% larger)

The screenshot displays the Ecoscope application interface. On the left is a sidebar with a menu containing: 'Set Workflow Details', 'Define time range', 'Data Source', 'Transform relocations to trajectories', and 'zoom to envelope'. The main panel on the right is titled 'Ecoscope' and contains several configuration sections. At the top, there are two date and time pickers: 'Since *' (set to 05/03/2026, 12:00 AM) and 'Until *' (set to 04/04/2026, 11:59 PM). Below these are three sections outlined with red boxes: 1. 'Data Source' section: Includes the instruction 'Select one of your configured data sources.' and a dropdown menu labeled 'Data Source*' with 'Amboseli Trust for Elephants' selected. 2. 'Transform relocations to trajectories' section: Features a collapsed dropdown menu labeled 'Advanced Configurations'. 3. 'zoom to envelope' section: Contains an 'Expansion Factor' input field with the value '1.05' and a descriptive text below it: 'Factor to expand the bounding box (e.g., 1.2 = 20% larger)'. At the bottom right of the main panel are two buttons: 'Cancel' and 'Submit'.

Data Source, Transform Relocations, and Zoom to Envelope

Running the Workflow

Once all parameters are configured, click **Submit**. The runner will:

1. Pull vehicle GPS observations from EarthRanger for the specified time range.
2. Download the static boundary files (group ranches, conflict hotspots).
3. Convert observations to relocations, then build trajectory segments with speed and distance metrics.
4. Classify speed into 6 equal-interval bins and generate the Speed Map.
5. Render the Vehicle Tracks map using a uniform blue path layer.
6. Draw the speed-over-time line chart per vehicle.
7. Compute per-vehicle summary statistics (mean, min, max speed; total distance).
8. Assemble the Word report (cover page + per-vehicle sections) and the dashboard.
9. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS` .

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/` :

File	Description
<code>vehicle_relocations.geoparquet</code>	Cleaned GPS fix locations
<code>vehicle_trajectories.geoparquet</code>	Trajectory segments with speed and distance
<code><vehicle>_speedmap.html</code>	Interactive speed map per vehicle
<code><vehicle>_tracks.html</code>	Interactive tracks map per vehicle
<code><vehicle>_speed_line_chart.html</code>	Interactive speed-over-time line chart

File	Description
<vehicle>_speedmap.png	Screenshot of speed map (2× resolution)
<vehicle>_tracks.png	Screenshot of tracks map (2× resolution)
<vehicle>_speed_line_chart.png	Screenshot of speed line chart (2× resolution)
<vehicle>_summary.csv	Speed and distance summary table with totals row
lg_cover_page.docx	Rendered report cover page
<vehicle>.docx	Per-vehicle report section
Merged report .docx	Final combined Word report